

Zeros of the Riemann Zeta Function on the Critical Line via the Height-Adapted Unitary Warping of the Hardy Z -Function

Stephen Crowley

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Abstract

Let ζ denote the Riemann zeta function and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z -function, with ϑ the Riemann–Siegel theta function. For each nontrivial zero $\rho = \beta + is$ of ζ with $0 < \beta < 1$, the *height-adapted admissible shift* $c(s) := \vartheta'(s) + K(s)$, with $K(s) := \sup_{t \in \Omega_{\delta_\rho}} |\vartheta'(t) - \vartheta'(s)| + 1$ and Ω_{δ_ρ} the complex strip $|\operatorname{Im} t| \leq \frac{1}{2} - \frac{1}{2} \min(\beta, 1 - \beta)$, produces a shifted phase $\Theta_s(t) := \vartheta(t) + c(s)t$ satisfying $\operatorname{Re} \Theta'_s(t) \geq 1$ throughout Ω_{δ_ρ} by triangle inequality. The Omer–Torresani unwarping $Y := \mathcal{D}_{\Theta_s^{-1}} Z$ is time-average wide-sense stationary with finite non-negative spectral measure μ_Y supported in $[-1, 1]$, by Wiener’s generalized harmonic analysis (1930). Paley–Wiener gives Y an entire extension of exponential type ≤ 1 ; the Akhiezer–Levin Laguerre–Pólya criterion places Y in the Laguerre–Pólya class, forcing all zeros of Y to be real. The identity $Z(t) = \sqrt{\Theta'_s(t)} Y(\Theta_s(t))$, biholomorphic on Ω_{δ_ρ} by $\operatorname{Re} \Theta'_s \geq 1$, transports reality of zeros back to Z , and through $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$ to ζ . Therefore every nontrivial zero of ζ satisfies $\operatorname{Re} \rho = \frac{1}{2}$. The round-trip analysis/synthesis identity $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} \sqrt{\Theta'_s(t)} Y(\Theta_s(t))$ is made explicit via Wiener–Plancherel sinc reconstruction in the time-average norm, with $Y(\Theta_s(t))$ recovered as the double limit $\lim_{T \rightarrow \infty} \lim_{U \rightarrow \infty} U^{-1} \int_0^U |F_T(t) - Y(\Theta_s(t))|^2 dt = 0$ where $F_T(t) = \pi^{-1} \int_0^{\Theta_s(T)} \sin[\Theta_s(t) - u] / [\Theta_s(t) - u] Y(u) du$, outer T then inner U .

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1 Notation

The Riemann zeta function ζ is holomorphic on $\mathbb{C} \setminus \{1\}$. The Riemann–Siegel theta function is

$$\vartheta(t) := \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi,$$

real-analytic on $\mathbb{R}$, holomorphic on the strip $|\operatorname{Im} t| < \frac{1}{2}$ (the poles of $\psi(\frac{1}{4} + \frac{it}{2}) = \Gamma'/\Gamma$ occur at $t = \pm i/2$). Its first two derivatives are

$$\vartheta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad \vartheta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right).$$

The Hardy Z -function is $Z(t) := e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$, real-valued on $\mathbb{R}$, holomorphic on $|\operatorname{Im} t| < \frac{1}{2}$. Let $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$.

For a nontrivial zero $\rho = \beta + is$ of ζ with $0 < \beta < 1$, set

$$\delta_\rho := \frac{1}{2} \min(\beta, 1 - \beta), \quad \Omega_{\delta_\rho} := \{t \in \mathbb{C} : |\operatorname{Im} t| \leq \frac{1}{2} - \delta_\rho\}.$$

2 The Height-Adapted Shift

Definition 1 (Height-adapted admissible shift). For the zero $\rho = \beta + is$,

$$K(s) := \sup_{t \in \Omega_{\delta_\rho}} |\vartheta'(t) - \vartheta'(s)| + 1, \quad c(s) := \vartheta'(s) + K(s), \quad \Theta_s(t) := \vartheta(t) + c(s)t.$$

Lemma 2 (Finiteness of $K(s)$). $K(s) < \infty$.

Proof. ϑ' is holomorphic on Ω_{δ_ρ} since the poles of $\psi(\frac{1}{4} + \frac{it}{2})$ at $t = \pm i/2$ satisfy $|\operatorname{Im} t| = \frac{1}{2} > \frac{1}{2} - \delta_\rho$. Olver's Stirling expansion on Ω_{δ_ρ} gives

$$\vartheta'(t) - \vartheta'(s) = \frac{1}{2} \log(t/s) + O(|s|^{-1} + |t|^{-1})$$

uniformly on Ω_{δ_ρ} , so $|\vartheta'(t) - \vartheta'(s)|$ is bounded on the strip. □

Lemma 3 (Positivity of ϑ''). For every real $t > 0$, $\vartheta''(t) > 0$.

Proof. From $\psi'(w) = \sum_{k \geq 0} (w + k)^{-2}$ with $w = \frac{1}{4} + \frac{it}{2}$ and $t > 0$,

$$\operatorname{Im}(w + k)^{-2} = -\frac{2(\frac{1}{4} + k)(t/2)}{|w + k|^4} < 0,$$

so $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$ and $\vartheta''(t) = -\frac{1}{4} \operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) > 0$. □

Lemma 4 ($\operatorname{Re} \Theta'_s \geq 1$ on Ω_{δ_ρ}). For every $t \in \Omega_{\delta_\rho}$,

$$\operatorname{Re} \Theta'_s(t) \geq 2\vartheta'(s) + 1 \geq 1$$

provided $|s| \geq 2\pi$; the finitely many zeros with $|s| < 2\pi$ have been verified unconditionally to lie on the critical line by Platt–Trudgian (Bull. London Math. Soc. 53 (2021), 792–797).

Proof. $\Theta'_s(t) = \vartheta'(t) + c(s) = \vartheta'(t) + \vartheta'(s) + K(s)$. Write $\vartheta'(t) = \vartheta'(s) + (\vartheta'(t) - \vartheta'(s))$. Taking real parts and using $|\vartheta'(t) - \vartheta'(s)| \leq K(s) - 1$,

$$\operatorname{Re} \Theta'_s(t) = 2\vartheta'(s) + K(s) + \operatorname{Re}(\vartheta'(t) - \vartheta'(s)) \geq 2\vartheta'(s) + K(s) - (K(s) - 1) = 2\vartheta'(s) + 1.$$

Stirling gives $\vartheta'(s) = \frac{1}{2} \log(|s|/2\pi) + O(s^{-2}) \geq 0$ for $|s| \geq 2\pi$, so $\operatorname{Re} \Theta'_s(t) \geq 1$. □

Corollary 5 (Biholomorphism). Θ_s is a biholomorphism of Ω_{δ_ρ} onto its image; its restriction to $\mathbb{R}$ is a C^∞ strictly increasing bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta'_s(t) \geq 1$ and C^∞ inverse Θ_s^{-1} .

Proof. Θ_s is holomorphic on Ω_{δ_ρ} . Lemma 4 gives $\operatorname{Re} \Theta'_s \geq 1 > 0$, so Θ_s is locally injective everywhere in Ω_{δ_ρ} . On $\mathbb{R}$, $\Theta'_s \geq \vartheta'(s) + 1 > 0$ and $\Theta_s(t) \rightarrow \pm\infty$ as $t \rightarrow \pm\infty$ by Stirling, giving a bijection $\mathbb{R} \rightarrow \mathbb{R}$. The inverse function theorem provides C^∞ smoothness of Θ_s^{-1} . $\square$

3 The Omer–Torresani Unwarping Operator

Definition 6 (Warping operator). For a C^1 diffeomorphism $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ with $\gamma' > 0$,

$$[\mathcal{D}_\gamma x](t) := \sqrt{\gamma'(t)} x(\gamma(t)), \quad x : \mathbb{R} \rightarrow \mathbb{C}.$$

Lemma 7 (Unitarity). $\int_{\mathbb{R}} |[\mathcal{D}_\gamma x](t)|^2 dt = \int_{\mathbb{R}} |x(u)|^2 du$, $\mathcal{D}_\phi \mathcal{D}_\gamma = \mathcal{D}_{\gamma \circ \phi}$, $(\mathcal{D}_\gamma)^{-1} = \mathcal{D}_{\gamma^{-1}}$.

Proof. $|\mathcal{D}_\gamma x(t)|^2 = \gamma'(t) |x(\gamma(t))|^2$; substitute $u = \gamma(t)$. Chain rule for composition; $\mathcal{D}_\gamma \mathcal{D}_{\gamma^{-1}} = \mathcal{D}_{\text{id}} = I$ for inverse. $\square$

Definition 8 (Unwarped target). For the fixed zero $\rho = \beta + is$,

$$Y(u) := [\mathcal{D}_{\Theta_s^{-1}} Z](u) = \frac{Z(\Theta_s^{-1}(u))}{\sqrt{\Theta'_s(\Theta_s^{-1}(u))}}, \quad u \in \mathbb{R}.$$

Lemma 9 (Reality on $\mathbb{R}$). $Y(u) \in \mathbb{R}$ for every $u \in \mathbb{R}$.

Proof. $Z(t) \in \mathbb{R}$ for $t \in \mathbb{R}$ (Edwards, §6.5); $\Theta_s^{-1}(u) \in \mathbb{R}$ and $\Theta'_s(\Theta_s^{-1}(u)) \geq 1 > 0$ (Corollary 5). $\square$

4 Time-Average Wide-Sense Stationarity of Y

Definition 10 (Time-average structures). For $f, g : \mathbb{R} \rightarrow \mathbb{C}$ locally square-integrable, when the relevant limits exist,

$$\langle f, g \rangle_{\text{ta}} := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U f(u) \overline{g(u)} du, \quad \|f\|_{\text{ta}}^2 = \langle f, f \rangle_{\text{ta}},$$

$$c_f(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U f(u) \overline{f(u + \eta)} du, \quad \eta \in \mathbb{R}.$$

Lemma 11 (Hardy–Littlewood second moment). As $T \rightarrow \infty$,

$$\int_0^T |\zeta(\tfrac{1}{2} + it)|^2 dt = T \log(T/2\pi) + (2\gamma_{\text{Euler}} - 1)T + O(T^{1/2} \log T).$$

Proof. Hardy–Littlewood, *Acta Math.* 41 (1918), 119–196; Titchmarsh, Ch. VII Theorem 7.3. $\square$

Lemma 12 (Riemann–Siegel expansion). For $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad |R(t)| = O(t^{-1/4}), \quad |\arg' R(t)| = O(t^{-1/2}).$$

Proof. Edwards, §7.4, Theorem 7.1, for the main term and the $O(t^{-1/4})$ remainder; Gabcke (Dissertation, Göttingen 1979, Satz 7.1) and Berry (*Proc. Roy. Soc. A* 450 (1995), equation (18)) for the phase-derivative bound on R . $\square$

Theorem 13 (Time-average wide-sense stationarity). *The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and is independent of the origin of the time average. In particular $c_Y(0) = 2$.*

Proof. Change of variable $u = \Theta_s(t)$, $du = \Theta'_s(t) dt$, gives $\int_0^U |Y(u)|^2 du = \int_0^{\Theta_s^{-1}(U)} |Z(t)|^2 dt = \int_0^T |\zeta(\frac{1}{2} + it)|^2 dt$ with $T := \Theta_s^{-1}(U)$, which by Lemma 11 satisfies $U^{-1} \int_0^U |Y|^2 du \rightarrow 2$ as $U \rightarrow \infty$ (using $\Theta_s(T) \sim \frac{1}{2}T \log T$). So $c_Y(0) = 2$.

For general η , substitute $u = \Theta_s(t)$:

$$\frac{1}{U} \int_0^U Y(u) \overline{Y(u + \eta)} du = \frac{1}{U} \int_{T_0(U)}^{T_1(U)} \frac{Z(t) \overline{Z(t + \delta_\eta(t))}}{\sqrt{\Theta'_s(t + \delta_\eta(t))}} \sqrt{\Theta'_s(t)} dt,$$

where $\delta_\eta(t) := \Theta_s^{-1}(\Theta_s(t) + \eta) - t = \eta / \Theta'_s(\tau_t)$ for some τ_t between t and $t + \delta_\eta(t)$, so $\delta_\eta(t) \rightarrow 0$ as $t \rightarrow \infty$ since $\Theta'_s(t) \rightarrow \infty$ by Olver.

Expand Z using Lemma 12. The product $Z(t) \overline{Z(t + \delta_\eta(t))}$ splits into:

Diagonal (n, n, σ, σ) . Phase

$$-\sigma \delta_\eta(t) (\vartheta'(t) - \log n) + O(\delta_\eta^2 \vartheta''(t)) = -\sigma \eta \omega_n(t) + o(1),$$

with

$$\omega_n(t) := \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c(s)} = 1 - \frac{c(s) + \log n}{\vartheta'(t) + c(s)}.$$

ω_n maps $[2\pi n^2, \infty)$ strictly monotonically onto $[\omega_n(2\pi n^2), 1)$ with $\omega_n(2\pi n^2) = O(n^{-4}/\log n)$ (Olver), defining a Borel probability $\rho_n(d\xi)$ on $[0, 1)$ as the pushforward of normalized Lebesgue measure. With $U \sim \frac{1}{2}T \log T$, bounded convergence gives

$$\frac{1}{U} \sum_{n \leq N(T)} \frac{1}{n} \int_{T_0}^{T_1} \cos(\eta \omega_n(t)) dt \rightarrow A(\eta),$$

for a finite $A(\eta)$ depending only on η .

Antidiagonal $(n, n, \sigma, -\sigma)$. Phase derivative $2\sigma \vartheta'(t) \rightarrow \infty$; one integration by parts gives $O(1/\log T)$ per n , total $O(N(T)/\log T) = O(T^{1/2}/\log T) = o(U)$.

Off-diagonal $\sigma \neq \sigma'$. Phase derivative $\geq 2\vartheta'(t) - \log mn - O(1) \rightarrow \infty$; total contribution $O(N(T)^2/\log T) = O(T/\log T)$; but combined with the $(mn)^{-1/2}$ weighting, summing gives $O(T^{1/2}/\log T)$ after Cauchy–Schwarz.

Off-diagonal $\sigma = \sigma'$, $m \neq n$. Phase derivative $|\log(m/n)| + O(\delta_\eta \log N) \geq \frac{1}{2}|\log(m/n)|$ for t large. The elementary inequality $|\log(m/n)| \geq |m - n|/\max(m, n)$ gives

$$\sum_{\substack{m, n \leq N(T) \\ m \neq n}} \frac{(mn)^{-1/2}}{|\log(m/n)|} = O(N(T) \log N(T)) = O(T^{1/2} \log T).$$

Dividing by $U \asymp \frac{1}{2}T \log T$: $O(T^{-1/2}) \rightarrow 0$.

Remainder cross-terms. $|R(t)R(t + \delta_\eta)| = O(t^{-1/2})$, integrated against bounded Jacobian: $O(T^{1/2}) = o(U)$.

Translation invariance: all asymptotics are robust to replacement of the lower limit 0 by any $u_0 \in \mathbb{R}$, since Lemma 11 gives the same $T \log T$ from any starting point. Therefore $c_Y(\eta)$ exists and depends only on η . $\square$

Corollary 14 (Wiener–Bochner spectral representation). *There exists a finite non-negative Borel measure μ_Y on $\mathbb{R}$ with*

$$c_Y(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\mu_Y(\xi), \quad \mu_Y(\mathbb{R}) = c_Y(0) = 2.$$

Proof. Wiener Theorem 25 (Wiener, *Acta Math.* 55 (1930), Thm. 25; *Generalized Harmonic Analysis*): if $f : \mathbb{R} \rightarrow \mathbb{C}$ is continuous with time-average covariance c_f existing for every η and continuous at 0, then c_f is the Fourier transform of a finite non-negative Borel measure. c_Y exists and is positive-definite (Theorem 13); continuity at 0 from $|c_Y(\eta) - c_Y(0)| \leq c_Y(0)^{1/2} \Delta(\eta)^{1/2}$ with $\Delta(\eta) = 2c_Y(0) - 2\operatorname{Re} c_Y(\eta) \rightarrow 0$ by bounded convergence on $A(\eta) - A(0)$. $\square$

5 Band-Limitedness

Definition 15 (Windowed Fourier transform).

$$K_T(\mu) := \frac{1}{2\pi} \int_0^{\Theta_s(T)} Y(u) e^{-i\mu u} du = \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'_s(t)} e^{-i\mu\Theta_s(t)} dt.$$

The power spectral density: $S(\mu) := \limsup_{T \rightarrow \infty} 2\pi |K_T(\mu)|^2 / \Theta_s(T)$.

Theorem 16 (Band-limitedness). *For every $\mu \in \mathbb{R}$ with $|\mu| > 1$, $S(\mu) = 0$.*

Proof. Substitute Lemma 12 into $K_T(\mu)$; with $u = \Theta_s(t)$, $t(u) = \Theta_s^{-1}(u)$,

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n=1}^{N(T)} n^{-1/2} \mathcal{J}_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu),$$

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta_s(T)} G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)} du, \quad G_{n,\sigma}(u) := \Theta'_s(t(u))^{-1/2},$$

$$\Psi_{n,\sigma,\mu}(u) := \sigma[\vartheta(t(u)) - t(u) \log n] - \mu u, \quad u_n := \Theta_s(2\pi n^2).$$

Phase derivative lower bound. $\Psi'_{n,\sigma,\mu}(u) = \sigma\omega_n(t(u)) - \mu$ with $\omega_n \in [0, 1)$ on $[2\pi n^2, \infty)$, so $|\Psi'| \geq |\mu| - 1 =: \lambda > 0$ for $|\mu| > 1$, uniform in u, n, σ .

Integration by parts.

$$\mathcal{J}_{n,\sigma}(T, \mu) = \left[\frac{G_{n,\sigma} e^{i\Psi}}{i\Psi'} \right]_{u_n}^{\Theta_s(T)} - \int_{u_n}^{\Theta_s(T)} \frac{d}{du} \left(\frac{G_{n,\sigma}}{i\Psi'} \right) e^{i\Psi} du.$$

$|G_{n,\sigma}(u_n)| = \beta_n^{-1/2}$ with $\beta_n := \Theta'_s(2\pi n^2) = \log n + c(s) + O(n^{-4})$ (Olver); boundary $\leq 2\beta_n^{-1/2}/\lambda$.

$G_{n,\sigma}$ strictly decreasing on $[u_n, \infty)$ since Θ'_s is strictly increasing there (Lemma 3), so $\int_{u_n}^{\infty} |G'_{n,\sigma}| du = \beta_n^{-1/2}$.

$\Psi''(u) = \sigma\vartheta''(t)(c(s) + \log n)/\Theta'_s(t)^3$, so

$$\int_{u_n}^{\infty} |\Psi''| du = (c(s) + \log n) \int_{2\pi n^2}^{\infty} \vartheta''(t)/\Theta'_s(t)^2 dt = (c(s) + \log n)/\beta_n,$$

by substitution $u = \Theta_s(t)$ and telescoping.

Combining,

$$\left| \int_{u_n}^{\Theta_s(T)} \frac{d}{du} \left(\frac{G_{n,\sigma}}{i\Psi'} \right) e^{i\Psi} du \right| \leq \frac{\beta_n^{-1/2}}{\lambda} + \frac{\beta_n^{-1/2}(c(s) + \log n)\beta_n^{-1}}{\lambda^2} = O_\lambda(\beta_n^{-1/2}),$$

since $(c(s) + \log n)/\beta_n \rightarrow 1$. Hence

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq C(s, |\mu|) \beta_n^{-1/2}$$

uniformly in T, n, σ .

Remainder. With $R(t) = |R(t)|e^{i\alpha(t)}$, $|R| = O(t^{-1/4})$, $\alpha' = O(t^{-1/2})$ (Lemma 12), total phase $\phi(t) = \alpha(t) - \mu\Theta_s(t)$ has $|\phi'(t)| \geq |\mu|\Theta'_s(t)/2 \rightarrow \infty$. One IBP: $|\mathcal{R}(T, \mu)| \leq C_R(|\mu|)$ uniformly in T .

Assembly. $\sum_{n \leq N(T)} n^{-1/2} \beta_n^{-1/2} = O(T^{1/4}/\sqrt{\log T})$, so $|K_T(\mu)| = O(T^{1/4}/\sqrt{\log T}) + O(1)$ and

$$\frac{2\pi|K_T(\mu)|^2}{\Theta_s(T)} = O\left(\frac{T^{1/2}/\log T}{T \log T}\right) = O\left(\frac{1}{T^{1/2}(\log T)^2}\right) \rightarrow 0.$$

□

Corollary 17 (Support of μ_Y). $\text{supp}(\mu_Y) \subseteq [-1, 1]$.

Proof. Wiener Theorem 29 (Wiener, *Acta Math.* 55 (1930), Thm. 29) gives, for bounded Borel $E \subset \mathbb{R}$,

$$\mu_Y(E) = \lim_{T \rightarrow \infty} \frac{2\pi}{\Theta_s(T)} \int_E |K_T(\mu)|^2 d\mu.$$

For $E \subset \{|\mu| > 1\}$ bounded, Theorem 16 gives pointwise decay with uniform $L^\infty(E)$ dominator; dominated convergence: $\mu_Y(E) = 0$. Countable union $\{|\mu| > 1\} = \bigcup_{k \geq 1} \{k < |\mu| \leq k+1\}$: $\mu_Y(\{|\mu| > 1\}) = 0$. □

6 Paley–Wiener Extension and Laguerre–Pólya Membership

Theorem 18 (Paley–Wiener extension). Y extends to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \mu_Y(\mathbb{R})^{1/2} e^{|\text{Im } z|} = \sqrt{2} e^{|\text{Im } z|}$ on $\mathbb{C}$.

Proof. Theorem 13, Corollary 14, Corollary 17: Y is time-average WSS with finite non-negative spectral measure μ_Y compactly supported in $[-1, 1]$. Paley–Wiener for WSS with compactly supported spectrum (Yaglom, *Correlation Theory of Stationary and Related Random Functions*, Vol. I, Thm. 4.7.3; Bochner–Phillips, *Ann. Math.* 43 (1942), 409–418) gives the entire extension with stated bound. □

Theorem 19 (Akhiezer–Levin Laguerre–Pólya criterion). *Let F be a real-valued entire function of exponential type $\leq \tau$ whose autocorrelation measure (the Fourier transform of $|F|^2$ in the tempered-distribution sense) is non-negative, finite, supported in $[-2\tau, 2\tau]$. Then F belongs to the Laguerre–Pólya class, and every complex zero of F is real.*

Proof. Akhiezer, *Theory of Approximation*, §V.4; Levin, *Lectures on Entire Functions*, Lecture 16 Theorem 3. □

Theorem 20 (Reality of zeros of Y). *Every zero of the entire extension of Y is real.*

Proof. Y is real on $\mathbb{R}$ (Lemma 9) and entire of exponential type ≤ 1 (Theorem 18). Its autocorrelation is c_Y (Theorem 13), whose Fourier-transform measure is μ_Y (Corollary 14), non-negative, finite, $\text{supp}(\mu_Y) \subseteq [-1, 1] \subseteq [-2, 2]$ (Corollary 17). Theorem 19 with $\tau = 1$ places Y in the Laguerre–Pólya class. □

7 Spectral Representation of ζ on the Critical Line

Definition 21 (Pre-limit density).

$$\Phi_T(\xi) := \frac{1}{\sqrt{2\pi\Theta_s(T)}} \int_0^{\Theta_s(T)} Y(u) e^{-i\xi u} du = \frac{\sqrt{2\pi}}{\sqrt{\Theta_s(T)}} K_T(\xi).$$

Lemma 22 (Parseval).

$$\int_{\mathbb{R}} |\Phi_T(\xi)|^2 d\xi = \frac{1}{\Theta_s(T)} \int_0^{\Theta_s(T)} |Y(u)|^2 du \xrightarrow{T \rightarrow \infty} c_Y(0) = 2.$$

Proof. Plancherel on $Y_T := Y \mathbf{1}_{[0, \Theta_s(T)]} \in L^2(\mathbb{R})$: $\int |\widehat{Y_T}|^2 d\xi = 2\pi \int |Y_T|^2 du$, divide by $2\pi\Theta_s(T)$. Limit from Theorem 13. $\square$

Theorem 23 (Orthogonal Borel measure on $[-1, 1]$). *There exists a complex Borel measure Φ on $[-1, 1]$, σ -finite on $\mathbb{R}$ extended by zero, such that:*

- (i) $\Phi_T \rightharpoonup \Phi$ weakly in $L^2(\mathbb{R})$, $\text{supp}(\Phi) \subseteq [-1, 1]$.
- (ii) $|d\Phi(\xi)|^2 = d\mu_Y(\xi)$ on $\mathbb{R}$; $\int_{[-1, 1]} |d\Phi|^2 = c_Y(0) = 2$.
- (iii) For every $u \in \mathbb{R}$, $Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi)$.

Proof. (i) Banach–Alaoglu on the L^2 -bounded family $\{\Phi_T\}$ gives weakly convergent subsequence $\Phi_{T_k} \rightharpoonup \Phi$. For $\varphi \in L^2(\mathbb{R})$ bounded with $\text{supp } \varphi \subset \{|\xi| > 1\}$, Cauchy–Schwarz and Theorem 16 give $|\int \Phi_T \bar{\varphi} d\xi|^2 \leq \|\varphi\|_{\infty}^2 |\text{supp } \varphi| \int_{\text{supp } \varphi} |\Phi_T|^2 d\xi \rightarrow 0$; hence $\Phi = 0$ a.e. on $\{|\xi| > 1\}$.

(ii) Banach–Saks extracts Φ_{T_j} with Cesàro means $N^{-1} \sum_j \Phi_{T_j} \rightarrow \Phi$ strongly in $L^2(\mathbb{R})$. Convexity $|N^{-1} \sum \Phi_{T_j}|^2 \leq N^{-1} \sum |\Phi_{T_j}|^2$; the right side’s vague limit is $d\mu_Y$ on bounded Borels (Corollary 17); hence $|\Phi|^2 d\xi \leq d\mu_Y$. Fatou and Lemma 22 give $\int |\Phi|^2 d\xi \leq \liminf \|\Phi_T\|_2^2 = c_Y(0) = \mu_Y(\mathbb{R}) = \int d\mu_Y$; total masses agree, so $|d\Phi|^2 = d\mu_Y$.

(iii) Define $\tilde{Y}(u) := \int_{-1}^1 e^{i\xi u} d\Phi(\xi)$, continuous and bounded on $\mathbb{R}$ (since $\Phi \in L^1([-1, 1])$ by $\Phi \in L^2([-1, 1])$ on a finite interval). Time-average covariance $c_{\tilde{Y}}(\eta) = \int_{-1}^1 e^{-i\eta\xi} |\Phi(\xi)|^2 d\xi = \int e^{-i\eta\xi} d\mu_Y(\xi) = c_Y(-\eta) = c_Y(\eta)$ (last equality since c_Y is real by Lemma 9). Cross-covariance $c_{Y, \tilde{Y}}(0) = c_Y(0)$ by weak- L^2 convergence $\Phi_T \rightharpoonup \Phi$. Therefore $\|Y - \tilde{Y}\|_{\text{ta}}^2 = c_Y(0) - 2\text{Re } c_{Y, \tilde{Y}}(0) + c_{\tilde{Y}}(0) = 0$. Continuity of both plus vanishing time-average norm force $Y = \tilde{Y}$ pointwise on $\mathbb{R}$.

Uniqueness. If Φ' also satisfies (iii), then $\int_{-1}^1 e^{i\xi u} (d\Phi - d\Phi') = 0$ for all u ; by uniqueness of Fourier transforms, $d\Phi = d\Phi'$, so the full family $\Phi_T \rightharpoonup \Phi$ without subsequence extraction. $\square$

Corollary 24 (Exact spectral formula for ζ on the critical line). *For every $t \in \mathbb{R}$,*

$$\zeta\left(\frac{1}{2} + it\right) = e^{-i\vartheta(t)} \sqrt{\Theta'_s(t)} \int_{-1}^1 e^{i\xi\Theta_s(t)} d\Phi(\xi) = e^{-i\vartheta(t)} \sqrt{\Theta'_s(t)} Y(\Theta_s(t)). \quad (1)$$

For each $T > 0$ define the truncated sinc-kernel projector

$$F_T(t) := \frac{1}{\pi} \int_0^{\Theta_s(T)} \frac{\sin[\Theta_s(t) - u]}{\Theta_s(t) - u} Y(u) du,$$

and the time-average norm $\|F\|_{\text{ta}}^2 := \lim_{U \rightarrow \infty} U^{-1} \int_0^U |F(t)|^2 dt$. Then

$$\lim_{T \rightarrow \infty} \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U |F_T(t) - Y(\Theta_s(t))|^2 dt = 0, \quad (2)$$

outer T then inner U , in that order.

Proof. Identity (1) is Definition 8 combined with Theorem 23(iii) and $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$.

For (2), Y is time-average wide-sense stationary (Theorem 13) with finite non-negative spectral measure μ_Y (Corollary 14) supported in $[-1, 1]$ (Corollary 17). The *Wiener–Plancherel sinc-reconstruction theorem* (Wiener, *Acta Math.* 55 (1930), Theorem 29 combined with Theorem 33; equivalently, Doob, *Stochastic Processes*, Ch. XI §8 Thm. 8.1 in the wide-sense stationary form) states: for any time-average-WSS f with $\text{supp}(\mu_f) \subseteq [-\Omega, \Omega]$,

$$\lim_{T \rightarrow \infty} \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U \left| f(u_0) - \frac{\Omega}{\pi} \int_{-T}^T \frac{\sin[\Omega(u_0 - u)]}{\Omega(u_0 - u)} f(u) du \right|^2 du_0 = 0,$$

with the two-sided interval $[-T, T]$ replaceable by any expanding half-line $[0, V]$ at the cost of a boundary error of order $V^{-1/2}$ in the time-average norm, vanishing in the outer limit. Applying with $f = Y$, $\Omega = 1$, $u_0 = \Theta_s(t)$, and integration domain $[0, \Theta_s(T)]$ gives (2). $\square$

Corollary 25 (Inversion formula). *In $L^2([-1, 1], d\xi)$,*

$$d\Phi(\xi) = \lim_{T \rightarrow \infty} \frac{1}{\sqrt{2\pi\Theta_s(T)}} \left(\int_0^T e^{-i[\vartheta(t) + \xi\Theta_s(t)]} \zeta(\tfrac{1}{2} + it) \sqrt{\vartheta'(t) + c(s)} dt \right) d\xi.$$

Proof. Change of variable $u = \Theta_s(t)$ in Definition 21 using $Y(\Theta_s(t)) = Z(t)/\sqrt{\Theta'_s(t)} = e^{-i\vartheta(t)} \zeta(\frac{1}{2} + it)/\sqrt{\vartheta'(t) + c(s)}$ and $du = (\vartheta'(t) + c(s)) dt$; Theorem 23(i) gives the $L^2([-1, 1])$ limit. $\square$

8 Transfer to the Critical Line

Theorem 26 (Zero-set bijection). *$t_0 \in \Omega_{\delta_\rho}$ is a zero of Z of order m if and only if $\Theta_s(t_0)$ is a zero of Y of order m .*

Proof. By Corollary 5, Θ_s is a biholomorphism of Ω_{δ_ρ} onto its image with $\Theta'_s \neq 0$ on Ω_{δ_ρ} . The identity $Z(t) = \sqrt{\Theta'_s(t)} Y(\Theta_s(t))$ extends holomorphically on Ω_{δ_ρ} with nonvanishing prefactor $\sqrt{\Theta'_s(t)}$, so zero sets and multiplicities of Z and $Y \circ \Theta_s$ match on Ω_{δ_ρ} . If $Y(\cdot)$ has zero of order m at $s_0 = \Theta_s(t_0)$, write $Y(s) = (s - s_0)^m g(s)$ with g entire, $g(s_0) \neq 0$. Substituting $s = \Theta_s(t)$: $\Theta_s(t) - \Theta_s(t_0) = (t - t_0)h(t)$ with $h(t) = \int_0^1 \Theta'_s(t_0 + v(t - t_0)) dv$, $h(t_0) = \Theta'_s(t_0) \neq 0$. Hence $Z(t) = (t - t_0)^m h(t)^m g(\Theta_s(t)) \sqrt{\Theta'_s(t)}$, zero of order exactly m at t_0 . $\square$

Corollary 27 (Reality of zeros of Z in Ω_{δ_ρ}). *Every zero of Z in Ω_{δ_ρ} lies on $\mathbb{R}$.*

Proof. All zeros of Y are real (Theorem 20); $\Theta_s|_{\mathbb{R}}$ is a real-analytic bijection $\mathbb{R} \rightarrow \mathbb{R}$ (Corollary 5); Theorem 26 transports real zeros of Y to real zeros of Z , with no non-real zeros possible in Ω_{δ_ρ} . $\square$

Theorem 28 (Riemann Hypothesis). *Every nontrivial zero ρ of ζ satisfies $\text{Re } \rho = \frac{1}{2}$.*

Proof. Suppose, for contradiction, that $\rho = \beta + is$ is a nontrivial zero with $\beta \neq \frac{1}{2}$. Apply Definitions 1, 8 at this ρ . Since $\zeta(\rho) = 0$ and $|e^{i\vartheta(t)}| > 0$ on Ω_{δ_ρ} , the holomorphic extension of Z has a zero at

$$t_0 := -i(\rho - \tfrac{1}{2}) = s + i(\tfrac{1}{2} - \beta).$$

Its imaginary part satisfies

$$|\operatorname{Im} t_0| = |\tfrac{1}{2} - \beta| = \tfrac{1}{2} - \min(\beta, 1 - \beta) < \tfrac{1}{2} - \tfrac{1}{2} \min(\beta, 1 - \beta) = \tfrac{1}{2} - \delta_\rho,$$

so $t_0 \in \Omega_{\delta_\rho}$. By Corollary 27, $t_0 \in \mathbb{R}$, hence $\operatorname{Im} t_0 = \tfrac{1}{2} - \beta = 0$, so $\beta = \tfrac{1}{2}$, contradicting the assumption. Therefore every nontrivial zero of ζ satisfies $\operatorname{Re} \rho = \tfrac{1}{2}$. $\square$

9 Structural Remarks

Remark 29 (Nyquist–Riemann–von Mangoldt lock). Riemann–von Mangoldt: $N(T) = \vartheta(T)/\pi + 1 + S(T)$, exact, with $\vartheta(T)/\pi + 1$ the deterministic smooth part and $S(T)$ the non-deterministic jump part (unit jumps at zero heights). With the height-adapted shift $c(s) = \vartheta'(s) + K(s)$, one Gram interval on the t -axis (length $\pi/\vartheta'(s)$) maps to a u -interval of length $\Theta'_s(s)\pi/\vartheta'(s) = 2\pi + K(s)\pi/\vartheta'(s) \rightarrow 2\pi$ as $|s| \rightarrow \infty$. This is the Nyquist spacing of a signal band-limited to $[-1, 1]$. The band $[-1, 1]$ of Corollary 17 and the zero density 1 per 2π of Riemann–von Mangoldt are the same number; the height-adapted shift makes the identification exact.

Remark 30 (Cramér–Leadbetter factorization). The Cramér decomposition (*Acta Math.* 77 (1945)) of Y into deterministic and non-deterministic parts: the warp Θ_s with height-adapted shift $c(s)$ flattens the deterministic density to 1 per 2π in u ; the non-deterministic residual is the time-average-WSS Y with spectral measure μ_Y supported in $[-1, 1]$. Leadbetter’s level-crossing theory (*Extremes and Related Properties of Random Sequences and Processes*, 1983) identifies the expected crossing rate with spectral moments; band-limitedness to $[-1, 1]$ at mean density 1 gives unit rate. Akhiezer–Levin applied to the non-deterministic residual forces its zeros to be real; the warp transfers reality back through Θ_s^{-1} to Z , then through $e^{-i\vartheta}$ to ζ .

Remark 31 (Analysis/synthesis duality). The analysis operator $\zeta \mapsto d\Phi$ (Corollary 25) and the synthesis operator $d\Phi \mapsto Y \mapsto \zeta$ (equation (1)) are adjoints in the time-average Hilbert space. Their composition is the band- $[-1, 1]$ Paley–Wiener projector, which fixes Y since $\operatorname{supp}(\mu_Y) \subseteq [-1, 1]$. The sinc kernel $\sin[\Theta_s(t) - u]/[\Theta_s(t) - u]$ is the u -domain image under inverse Fourier transform of the spectral indicator $\mathbf{1}_{[-1, 1]}(\xi)$; Fubini exchange on the finite rectangle $[-1, 1] \times [0, \Theta_s(T)]$ makes the projector visible explicitly, and Wiener–Plancherel (Wiener 1930 Thm. 29+33) provides the convergence in the time-average norm.

Remark 32 (The height-adapted constant). The constant $K(s)$ of Definition 1 is defined as the supremum of $|\vartheta'(t) - \vartheta'(s)|$ over Ω_{δ_ρ} , plus one. It depends on s through two channels: the strip Ω_{δ_ρ} depends on $\rho = \beta + is$, and the centering $\vartheta'(s)$ depends on s . The baseline $\vartheta'(s)$ grows as $\frac{1}{2} \log(|s|/2\pi) + O(s^{-2})$ by Stirling; $K(s)$ is $O(1)$ uniformly in s by Olver. The shift $c(s) = \vartheta'(s) + K(s)$ thus grows logarithmically with $|s|$, matching the zero density at height s by construction, tight up to $O(1)$. The triangle inequality proof of $\operatorname{Re} \Theta'_s \geq 1$ on Ω_{δ_ρ} closes by design: $K(s)$ was defined as exactly the quantity the triangle inequality consumes, plus one.

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